

Natural Language
Processing,
Large Language
Models, Large
Multimodal Models,
Pretrained Models,
LLMs, LangChain,
Hugging Face



```
### (1) Packages
import os
from langchain.document_loaders import ...

### (2) File Handling
# Get the current working directory
file_path = os.path.abspath(__file__)
current_dir = os.path.dirname(file_path)

# Go up one directory level
parent_dir = os.path.dirname(current_dir)

file_path = os.path.join(parent_dir, 'data')
file_path
```

Generative AI with Python

The Developer's Guide to Pretrained LLMs, Vector Databases,
Retrieval-Augmented Generation, and Agentic Systems

Bert Gollnick

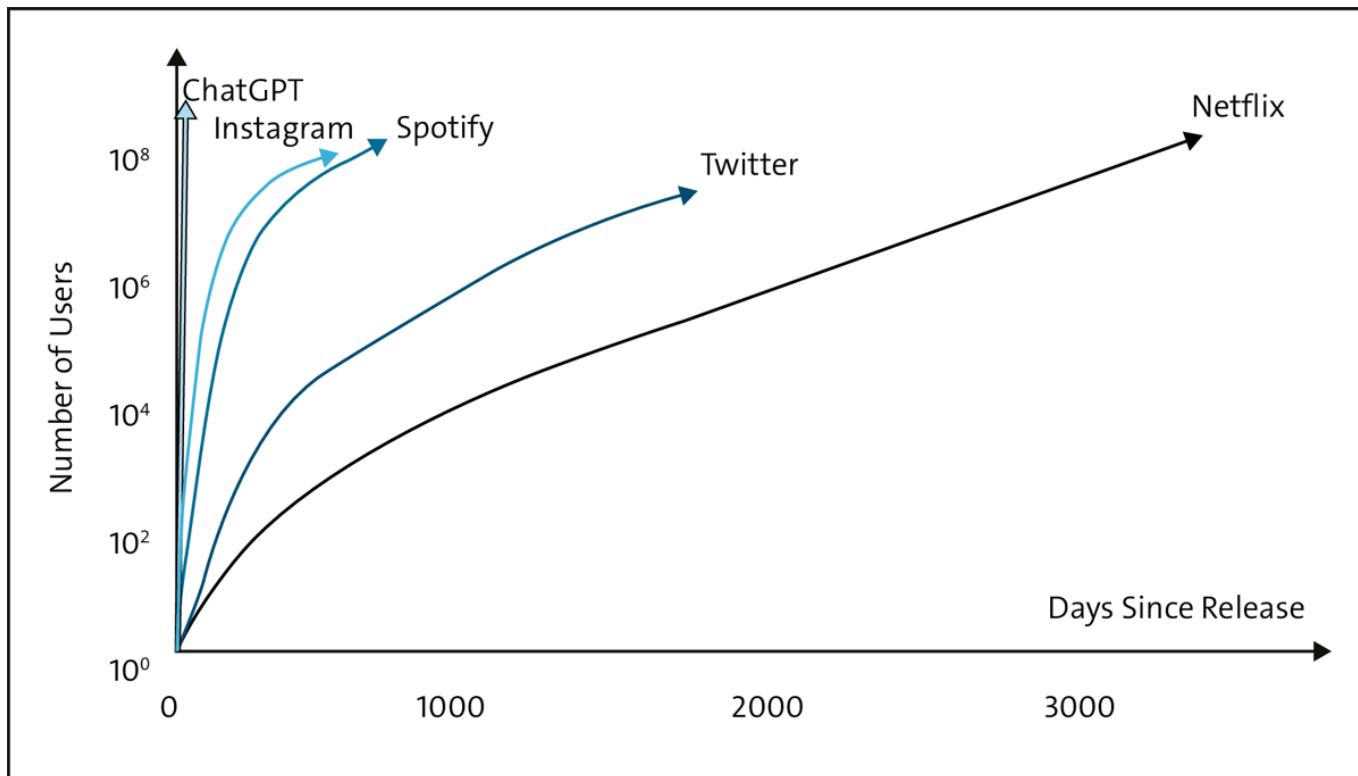
 Rheinwerk
Computing

Chapter-1

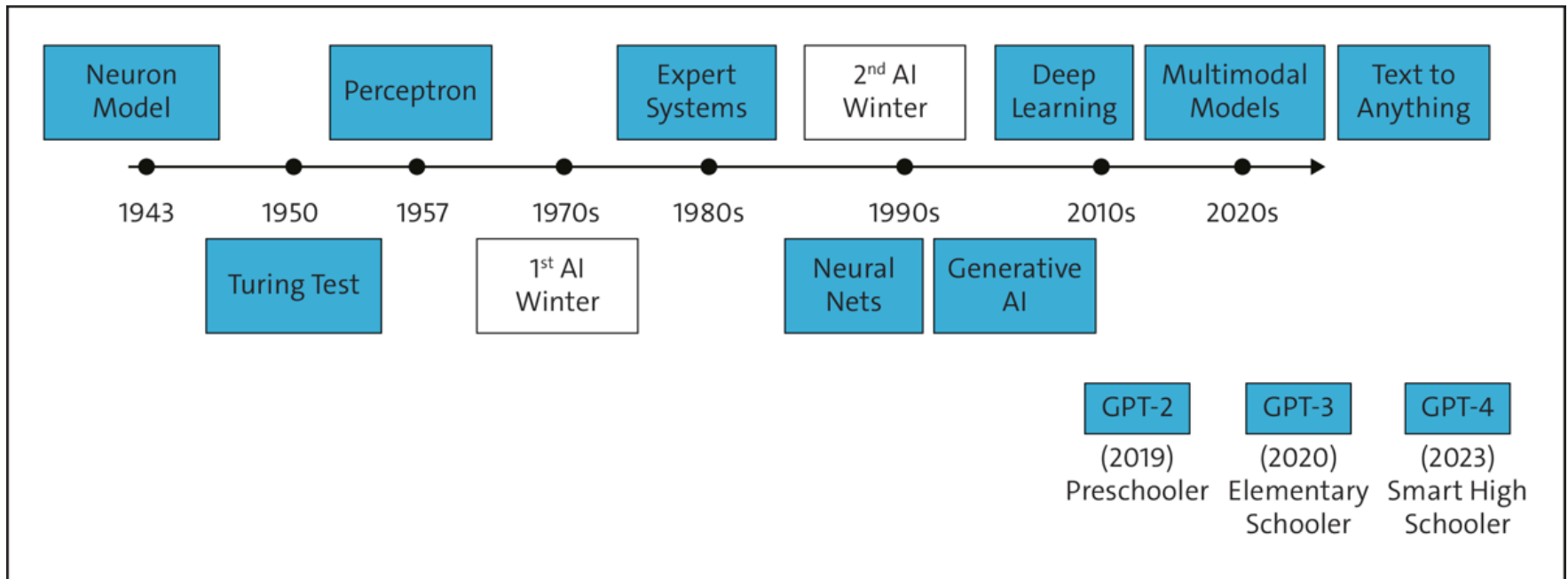
Introduction to Generative AI

1.1 Introduction to Artificial Intelligence

Rise of ChatGPT



Evolution of AI and it's importance

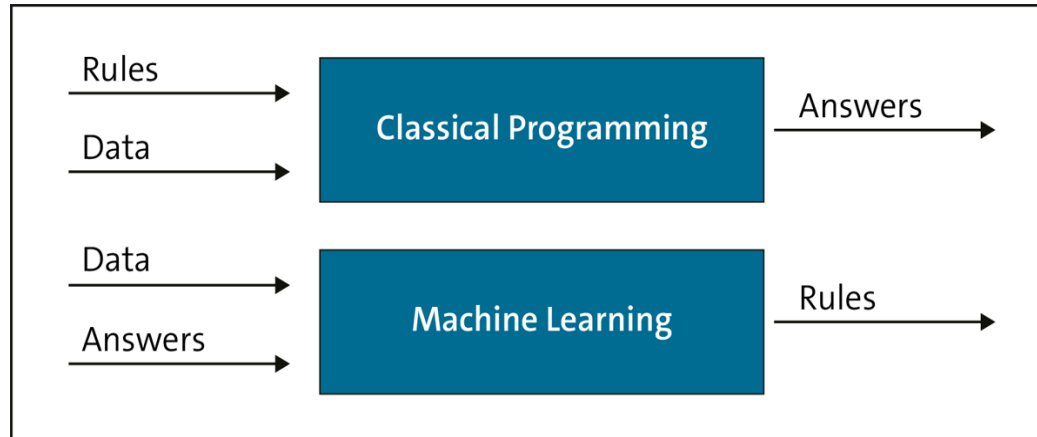


1.2 Pillars of Generative AI Advancement

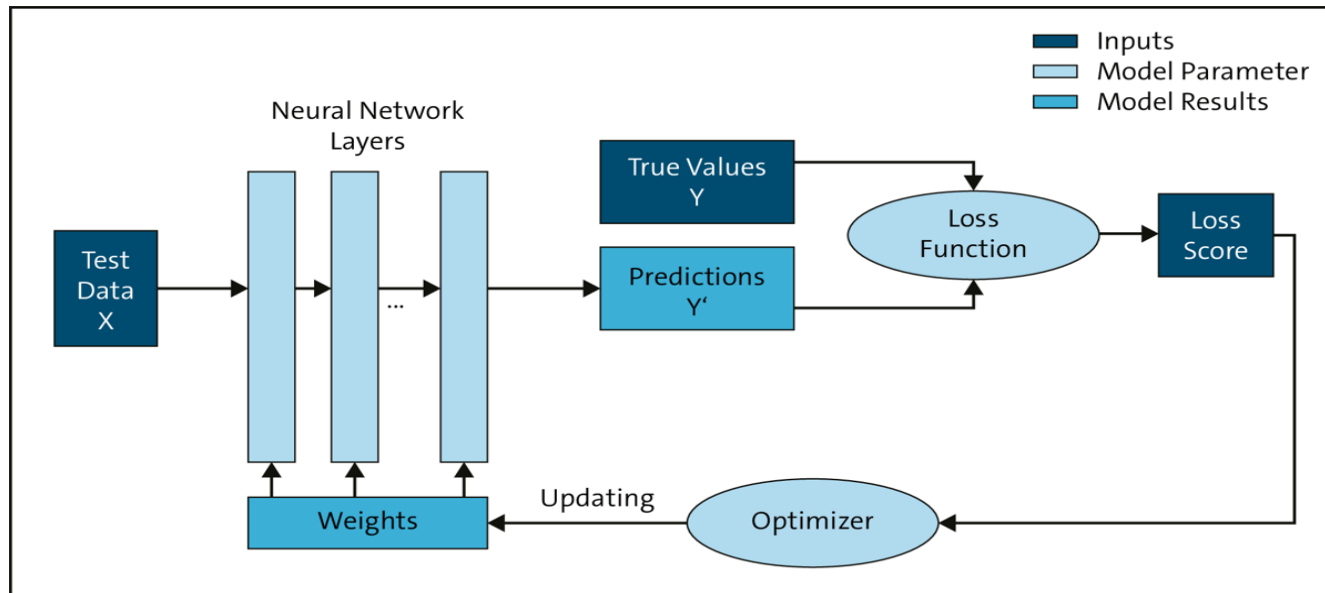
- Computational Power
- Model and Data Size
- Investments
- Algorithmic Improvements

1.3 Deep Learning

Traditional Programming vs Machine Learning

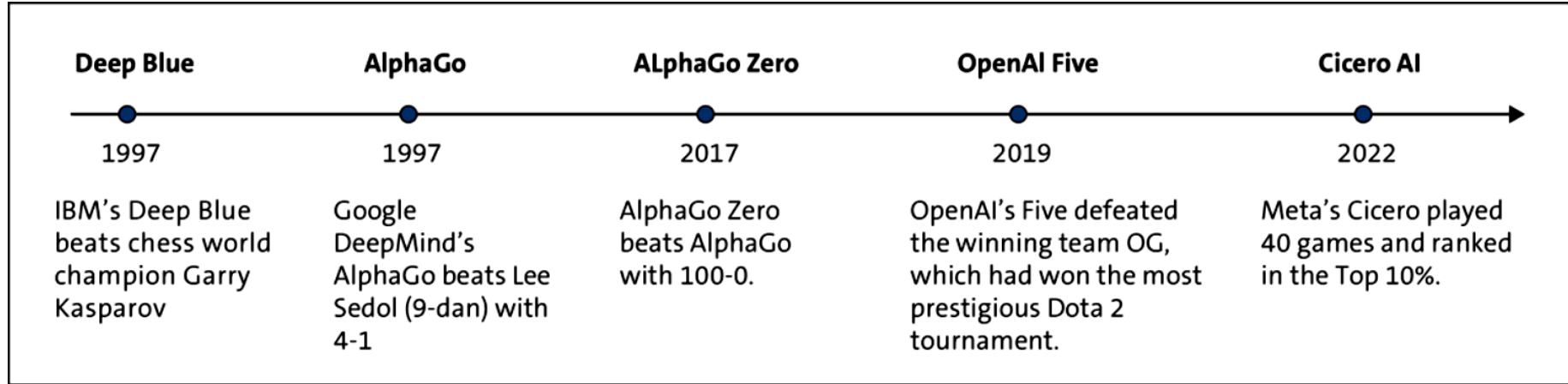


Deep Learning: Model Training



1.4 Narrow AI

Narrow AI Achievements



1.5 Natural Language Processing Models

- What is NLP?
- NLP Tasks



Text Classification



Text Generation



Question/Answering



Text Translation



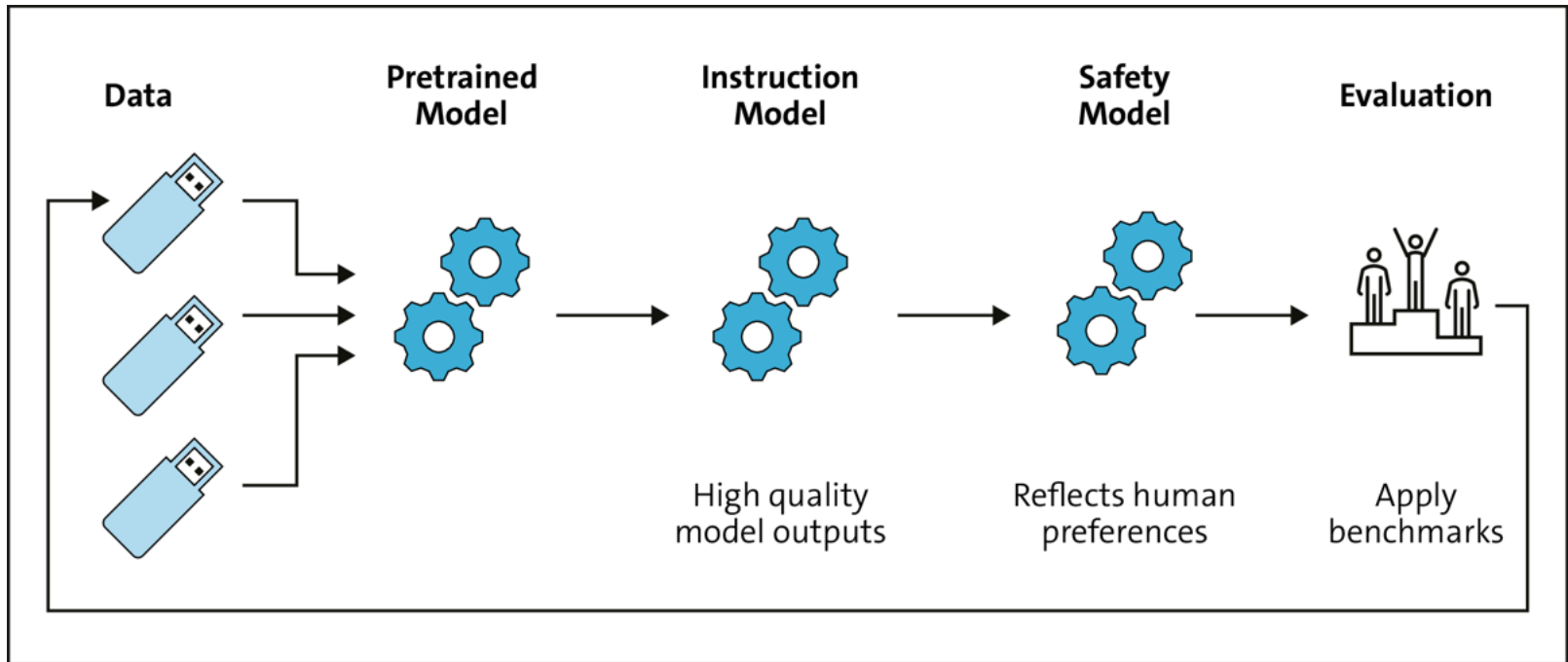
Fill-Mask



Sentence Similarity

1.6 Large Language Models

LLM Training Process



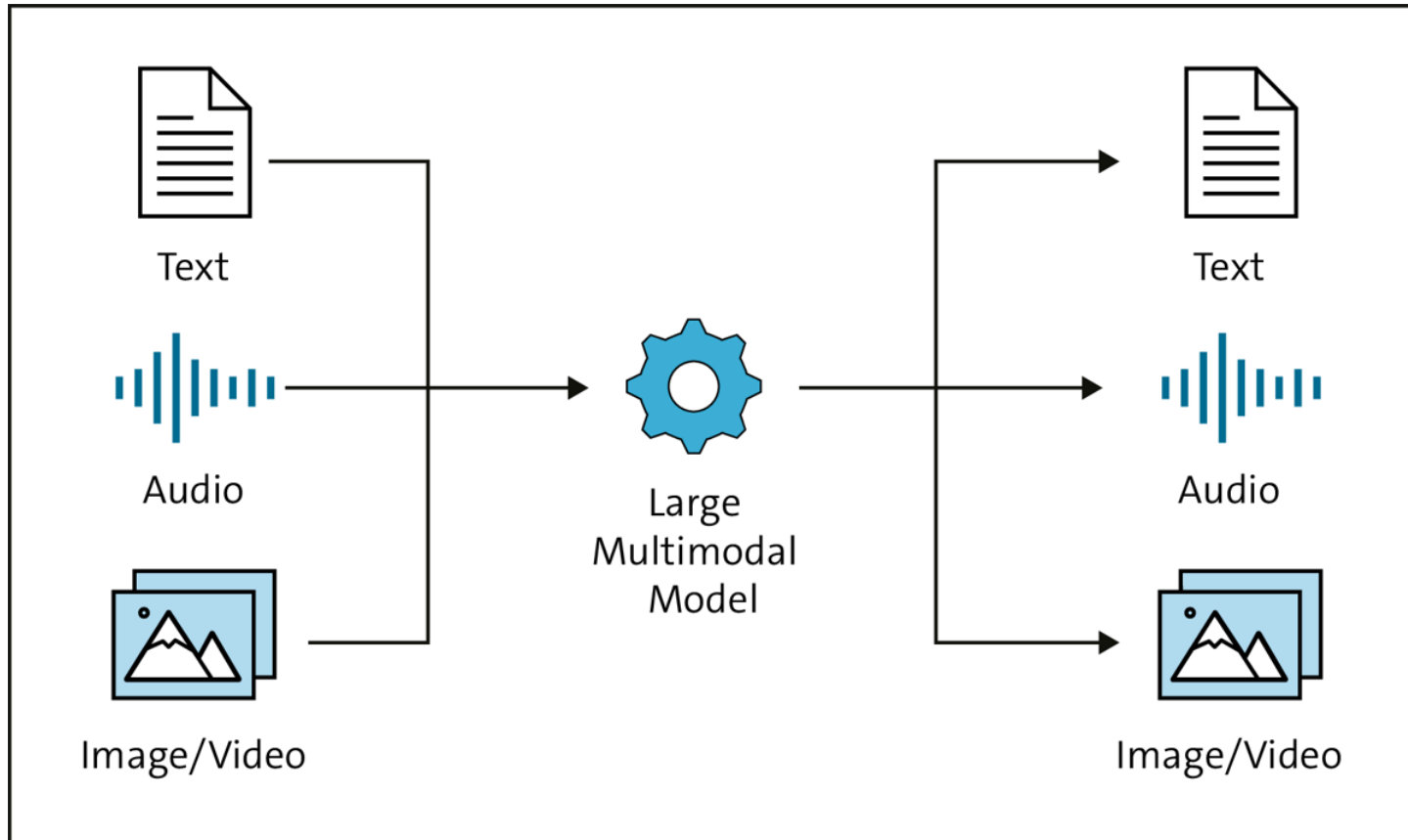
Use Cases of LLMs

- Creative Writing
- Summarization
- Chatbots and Virtual Assistants
- Translation
- Content Creation
- Education

Limitations of LLMs

- Hallucinations
- Bias
- Context Window and Token Limitations

1.7 Large Multimodal Models (LMMs)



1.8 Generative AI Applications

- Consumer Applications
- Enterprise – Horizontal Applications
- Enterprise – Vertical Applications
- Prosumer Applications